



The effect of pre-softened wood chips on wood fibre aspect ratio and mechanical properties of wood–polymer composites

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ABSTRACT

The objective of this work was to study the effect of chemical pre-treatment and moisture content of wood chips on the wood particle aspect ratio after compounding in a twin-screw extruder and on the mechanical properties of wood–polymer composites (WPCs). Composites with 50 wt.% wood content were manufactured using pre-treated and untreated wood chips. The effect of wood moisture content on composite properties was studied by using dried and undried wood chips. The mechanical properties and fracture surfaces of the composites as well as the microstructure and aspect ratio of wood particles after compounding were studied. The highest wood particle aspect ratio after extrusion was achieved by using pre-treated, undried wood chips as raw material. The chemical pre-treatment was found to enhance the defibration of wood chips as well as the mechanical properties of the composites.

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1. Introduction

Wood–polymer composites (WPCs) are materials which combine the properties of wood and thermoplastic polymers. They are considered more environmentally friendly options for wood and plastic products in several applications, such as outdoor building materials, furniture and interior panels in automobiles [1]. WPCs are manufactured by compounding wood raw material with a molten matrix polymer together with additives such as lubricants, antioxidants and coupling agents in various compositions. In commercial manufacturing of wood–polymer composites the main wood raw material used is wood flour. Wood flour is typically produced from the residues of wood processors by milling and screening the material into a specific particle size range [2]. Usually, the aspect ratio (length-to-diameter ratio) of wood flour particles is low, between 1 and 5 [2]. Earlier studies have shown that wood and other natural fibres with higher aspect ratio give better reinforcement than wood particles with low aspect ratio [3,4]. Especially when there is a good adhesion between fibres and the matrix, a better stress transfer to the fibres is achieved [2]. However, since the processing of longer, bulky fibres is more difficult and fibres are more expensive than wood flour, wood fibres are not often used in the manufacture of wood–polymer composites [5].

The focus in wood–polymer composite research has been on the improvement of the interfacial adhesion between the hydrophilic wood and hydrophobic polymer matrix. Studies have concentrated especially on various coupling agents and chemical treatments to improve adhesion between these two components [6–10]. Considerably less research has been focused on studying the effects of wood particle properties on composite properties, and these studies are mainly about how the wood particle size and wood species affect composite properties [11–14]. Only a few studies can be found where the effect of the aspect ratio of wood fillers has been addressed [15–17], even though it has been suggested that it is the aspect ratio but not the particle size of wood filler which has the main influence on the mechanical properties of the wood–polymer composites [4]. The possibility to use larger wood particles as raw material for WPC has been studied by only a few researchers [18,19], though it would be an economic advantage if wood residues could be used directly in the compounding process of wood–polymer composites without pre-processing steps such as milling, sieving and drying. In our earlier study it was shown that wood chips can be used directly as raw material for WPC, and it is even possible to separate individual fibres from the wood chips during the extrusion compounding if undried wood chips are used as raw material [20].

The objective of this work was to investigate the effect of chemical pre-treatment and moisture content of wood chips on the wood particle aspect ratio after compounding in a twin-screw extruder and on the mechanical properties of prepared composites. The purpose of the chemical pre-treatment was to soften the wood

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raw material prior to extrusion. In the earlier study by Hietala et al. [20] a decrease in the flexural properties was noticed when undried wood chips were used as raw material, even though the aspect ratio of wood particles was increased. The high moisture content (40 wt.%) of the wood raw material feed was believed to be the reason for this and therefore, in this work the effect of lower moisture content of the undried wood raw material was also studied.

2. Experimental

2.1. Materials

Spruce chips were used as wood raw material (Fig. 1). Wood chips were used as: (1) untreated and undried (UU), (2) untreated and dried (UD), (3) pre-treated and undried (PU), and (4) pre-treated and dried (PD). The moisture content of the undried wood chips was 33 wt.% based on total weight, and the moisture content of the dried wood chips was below 1 wt.% based on the dry weight. The wood chips were dried in an oven at 105 °C for 24 h.

The matrix polymer used was polypropylene, PP, in powder form with a Melt Flow Index (MFI) of 32 g/10 min, 230 °C/2.16 kg and 16 g/10 min, 200 °C/2.16 kg supplied by Borealis Polyolefins, Schwechat, Austria. A coupling agent, maleic anhydride grafted polypropylene, (MAPP, Epolene E-43, Eastman Chemical Company, Longview, Texas, USA), a lubricant (TPW 113, Struktol, Stow, Ohio, USA) and antioxidant agent were used as additives.

2.2. Wood chip pre-treatment

Sodium sulphite, Na₂SO₃ (Merck KGaA, Darmstadt, Germany), was used in the chemical pre-treatment. The pre-treatment was done at 90 °C and at pH 9 for 210 min. The wood raw material was diluted to 10% consistency and 5% Na₂SO₃ (based on the total mass of the mixture) was used in the treatment. After the pre-treatment, wood chips were washed with water, and then left to dry at room temperature until a moisture content of 33 wt.% was reached.

2.3. Compounding process

A co-rotating twin-screw extruder with twin-screw side feeder, ZSK-18 MEGALab (Coperion W&P, Stuttgart, Germany) equipped with a K-Tron gravimetric feeder (Niederlenz, Switzerland), two atmospheric vents and a vacuum vent was used to compound the composite materials. Polypropylene and the additives were pre-mixed prior to extrusion. The gravimetric feeder was used to feed PP into the extruder, while wood chips were fed into the extruder manually, because it was not possible to use the gravimetric

feeder system for this kind of material. The wood content of composites was calculated according to the following formula:

$$\text{Wood content, \%} = \frac{m_{\text{wood}}}{m_{\text{composite}}} \times 100\% = \frac{m_{\text{composite}} - (Q_{\text{PP}} \times t)}{m_{\text{composite}}} \times 100\% \quad (1)$$

where m_{wood} and $m_{\text{composite}}$ are the masses of wood (total mass) and composite (kg), Q_{PP} is the feeding rate of PP(kg/h), and t is the sampling time (h) used to collect the composite. The aim was to prepare composites with a wood content of 50 wt.%, but the actual wood contents varied between 45 wt.% and 54 wt.%.

The extrusion set-up and the screw configuration used in the compounding of the composites are shown in Fig. 2. The extrusion conditions were the same for all prepared composites. The screw configuration used was chosen based on the reported results of an earlier study [20]. The compounded composites were profiled into rectangular profile with a cross section of 5 mm × 20 mm. Four types of wood chip-PP composites were manufactured:

- (a) **UD-PP**: composite made with untreated and dried wood chips.
- (b) **UU-PP**: composite made with untreated and undried wood chips.
- (c) **PD-PP**: composite made with pre-treated and dried wood chips.
- (d) **PU-PP**: composite made with pre-treated and undried wood chips.

Except for the manual feeding of the wood chips, there were no great difficulties in the processing of the composites. Steam was created during compounding when undried wood chips were used, but it was ventilated out from the extruder. Blockage of the atmospheric vents after a longer compounding period posed a minor problem. The overall impression was that using undried wood chips was less detrimental to the extruder screws than the use of dried wood chips.

The appearance of the composite profiles made using undried wood chips was slightly different from the profiles made with dried wood chips indicating that there could be more voids in the material due to vaporized moisture. However, after compression moulding the specimens appeared to be identical.

2.4. Characterization

2.4.1. Specimen preparation

Test specimens for flexural testing were moulded with a conventional compression moulding press (Fjellman Press AB, Mariestad, Sweden). Before moulding, the extruded profiles were broken into shorter pieces (approximately 60 mm × 5 mm × 20 mm) and weighed into batches of ~34 g which were then heated between aluminium plates and Teflon sheets in a hot air oven at 200 °C for 20–45 min. After heating, the composites were transferred into the compression moulding press. The mould temperature was 50 °C and the pressure applied on the samples was approximately 70 MPa. A plate of five flexural specimens could be moulded at once. After cutting, the size of the flexural specimens was 5 mm × 10 mm × 120 mm.

Plate-shaped test specimens for impact testing were made with an LPC-300 Fontijne Grotnes heat press (Vlaardingen, Netherlands) and using a metal frame with an opening size of 1.5 mm × 100 mm × 100 mm. Before pressing, the extruded material was broken into shorter pieces (approximately 90 mm × 5 mm × 20 mm, five pieces per plate), placed between aluminium plates and Teflon sheets with the frame, and preheated in

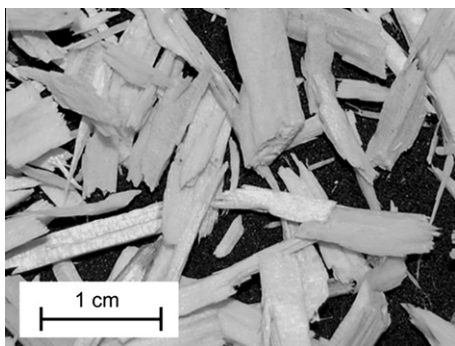


Fig. 1. Spruce wood chips.

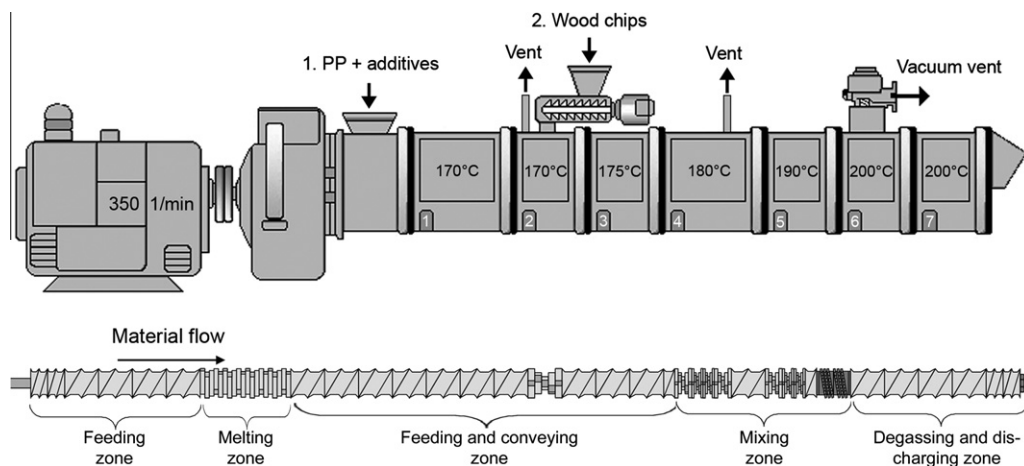


Fig. 2. The extrusion set-up and the screw configuration used in the study.

the press for 10 min at 200 °C without pressure. The preheated material was then pressed for 2 min, with a load of 150 kN at 200 °C. The dimensions of the test plates were approximately 1.6 mm × 100 mm × 100 mm.

2.4.2. Mechanical properties

Flexural testing of the wood chip-PP composites was performed according to ASTM D790 standard (three-point bending). A Shimadzu Autograph AG-X universal testing machine (Shimadzu Corp, Kyoto, Japan) with a load cell of 1 kN was used for testing. A minimum of 13 specimens of each material were tested. The support span used in the testing was 84.5 mm and the speed of the crosshead was 2.254 mm/min. Impact properties of the prepared composites were measured with a falling weight impact testing machine, Dynatup Minitower (Instron, Norwood, MA, USA) according to ASTM D3763 standard. The total mass of the impactor was 2.31 kg and speed of testing was 2.8 m/s. At least three specimens of each material were tested.

Because there were differences in the wood contents of the composite materials, the results from the flexural testing were normalized to correspond to 50 wt.% wood content assuming that linear relationship between wood loading and properties exists. The normalization was done using the following formula:

$$\text{Normalized value} = \text{Actual value} \times \frac{w_{\text{normalization}}}{w_{\text{specimen}}} \quad (2)$$

where $w_{\text{normalization}}$ is the chosen wood content for normalization (50 wt.%) and w_{specimen} is the actual wood content of the specimen (wt.%). For example, the flexural strength value of UD-PP specimen No. 8 was 46.11 MPa with a wood content of 54 wt.%. The normalized strength value was calculated using Eq. (2) as follows:

$$\begin{aligned} \text{Normalized flexural strength} &= 46.11 \text{ MPa} \times \frac{50 \text{ wt.}\%}{54 \text{ wt.}\%} \\ &= 42.69 \text{ MPa.} \end{aligned}$$

2.4.3. Density

The density of the compression moulded composite specimen was measured using the water displacement method according to ASTM D792 standard (test method A). The specimen size was 65 mm × 10 mm × 5 mm. Four replicates for each material were tested.

2.4.4. Extraction of matrix polymer

To study the wood particle size and shape after compounding, PP matrix was extracted from the composites. Prior to extraction, pieces of each composite material were pressed into thin sheets with a heat press (LPC-300, Fontijne Grotnes, Vlaardingen, Netherlands) by first preheating the material for 5 min without pressure at 200 °C and then pressing for 2 min at 150 kN at 200 °C. A few grams of each pressed material were kept in boiling xylene until the matrix was completely dissolved (24 h). After extraction, the wood particles were washed with ethanol and water to remove xylene.

2.4.5. Aspect ratio of wood particles

For the determination of the aspect ratio of wood particles extracted from the composites, the samples were first analyzed with a tube flow fractionator (Metso Automation, Kajaani, Finland). The tube flow fractionator is typically used to fractionate wood fibre samples into different size categories. However, in this case, the device was used to capture images of the wood particle samples for the image analysis, because during analysis approximately 600 images of the sample are recorded. The tube flow fractionation method is described in more detail in publications by Krogerus et al. and Laitinen et al. [21,22].

After fractionation the images were analyzed using kajaaniIMG image analysis software supplied by Metso Automation (Kajaani, Finland). During image analysis, captured images are segmented to regions that may have one or several objects. The objects (particles, fines and fibres) in one region are then interpreted based on geometrical criteria. Several image analysis algorithms are performed to measure the properties of each object. The basis for these algorithms is the model on interaction of light and various objects. This enables the use of so-called sub-pixel algorithms to improve the resolution. For each of the objects, the projection area is determined. Based on the shape of the object, other parameters such as length and width are then decided. The aspect ratio of the wood particles was calculated using the length and width parameters. The settings used in the image analysis are presented in Table 1. The settings were chosen so that the smallest wood particles would be excluded from the analysis.

2.4.6. Morphology and microstructure

A scanning electron microscope (Jeol JSM-6460, Jeol Ltd., Tokyo, Japan) was used to observe the size and shape of wood particles after extraction of the polymer matrix as well as the fracture surfaces of the manufactured composites. The wood particle samples

Table 1
Settings used in the image analysis.

Parameter	Value
Minimum fibre width	10 μm
Maximum fibre width	3000 μm
Minimum fibre length	0.1 mm
Maximum fibre length	7.6 mm

Table 2
Material compositions of the wood chip-PP composites.

Material	Weight (%)			
	PP ^a	Lubricant	MAPP	Wood
UD-PP	41.4	2.8	1.8	54.0
UU-PP	44.1	2.9	2.0	51.0
PD-PP	45.0	3.0	2.0	50.0
PU-PP	49.5	3.3	2.2	45.0

PP = polypropylene, UD = untreated and dried wood chips, UU = untreated and undried wood chips, PD = pre-treated and dried wood chips, PU = pre-treated and undried wood chips.

^a Polypropylene includes 0.18 wt.% of antioxidant agent.

were freeze-dried before microscopy. The fracture surfaces of the composites were created by freezing flexural testing specimens in liquid nitrogen and then breaking the frozen specimen. All the specimens were sputter-coated with gold before observation. An acceleration voltage of 10 kV was used in the observation of the fracture surfaces and an acceleration voltage of 5 kV was used in the observation of the wood particles.

2.4.7. Statistical analysis

One-way analysis of variance (ANOVA) followed by Tukey–Kramer multiple comparison tests with a 0.05 significance level was used to analyze the results from mechanical testing.

3. Results and discussion

3.1. Composites

In Table 2 the compositions of the wood chip-PP composites are shown. The wood contents of the composites varied to some extent due to the manual feeding of the wood chips in the extruder. In general, the composites with dried wood chips had higher wood content compared to the composites with undried wood.

3.1.1. Mechanical properties

The results from the mechanical testing are shown in Table 3. Because there were differences in the wood contents of the composite materials, the results from flexural testing were normalized to correspond to 50 wt.% wood content.

From Table 3 it can be seen that the chemical pre-treatment affected the flexural properties positively: the composites with pre-

treated wood chips had better strength properties than composites with untreated wood chips and the same wood chip moisture levels. The flexural properties on the composites were also affected by the moisture content of the wood chips: the use of undried wood chips with a moisture content of 33 wt.% enhanced the flexural strength but, simultaneously, the modulus of elasticity was decreased. The composite manufactured with pre-treated, undried wood chips (PU-PP) had the highest flexural strength and the composite with pre-treated, dried wood chips (PD-PP) had the highest flexural modulus among the prepared composites. The lower modulus of the composites with undried wood chips could be explained by the fact that moisture makes wood more flexible. From Table 3 it can be seen that the strains of the composites with undried wood chips were larger than with the composites with dried wood chips.

The impact properties of the composites were studied using a falling weight impact test, and the results are presented in Table 3. Characteristic load–deflection and energy–deflection curves from the impact testing are shown in Fig. 3. The shape of the curves was similar for all the materials. It can be seen from Table 3 that there were small differences between the peak load and total impact energy values of the composites; however, the differences were not statistically significant.

The results from the mechanical testing showed that the chemical pre-treatment enhanced the flexural properties of the composites. The flexural strength of the composites with pre-treated wood chips was higher than that of the composites with untreated chips. No significant improvement could be seen in the impact properties due to the chemical pre-treatment. The enhancement of the flexural strength of the composites with pre-treated wood chips is believed to be caused mainly by the higher aspect ratio of wood fibres and particles due to lignin softening. This is assumed to be especially pronounced when undried wood chips are used. It is also possible that pre-treatment with sodium sulphite changed the surface properties of the wood particles [23], and thus improved the mechanical properties of the composites by improving the adhesion between wood and polypropylene.

3.1.2. Density

The results from the density measurements are shown in Fig. 4. The results were normalized to correspond to 50 wt.% wood contents. The composite with pre-treated, undried chips (PU-PP) had the highest density, 1.23 g/cm³, and the composite with untreated, dried chips had the lowest density, 1.03 g/cm³. According to Clemons 2008 [2], increased density of the wood–polymer composites is often related to the collapsing or filling of the hollow wood fibres. The reason for the higher density of the PU-PP composite is most likely that there are more individual, collapsed wood fibres in comparison with the larger wood particles in the other composites due to the softening of the wood caused by moisture and the chemical pre-treatment. The composite with the highest density (PU-PP) had the best flexural properties among the prepared materials.

Table 3

Averages and standard deviations of flexural and impact properties of the composites. The flexural properties were normalized to correspond to 50 wt.% wood content.

Material	Normalized flexural properties			Impact properties	
	Strength* (MPa)	MOE* (GPa)	Strain at max* (%)	Peak load* (N)	Total energy* (J)
UD-PP	43.6 ± 3.9 ^a	4.5 ± 0.4 ^{ac}	1.4 ± 0.1 ^a	148.6 ± 10.2 ^a	1.9 ± 0.1 ^a
UU-PP	48.5 ± 4.5 ^b	3.9 ± 0.5 ^b	2.1 ± 0.2 ^b	167.6 ± 3.6 ^a	2.1 ± 0.4 ^a
PD-PP	50.5 ± 2.6 ^b	4.9 ± 0.5 ^a	1.6 ± 0.2 ^a	156.3 ± 13.6 ^a	1.7 ± 0.4 ^a
PU-PP	57.7 ± 5.4 ^c	4.3 ± 0.3 ^{bc}	2.5 ± 0.2 ^c	169.9 ± 25.8 ^a	2.0 ± 0.3 ^a

* Means marked with the same superscript letter within the same column are not significantly different at 5% significance level based on the ANOVA and Tukey–Kramer pairwise comparison test. PP = polypropylene, UD = untreated and dried wood chips, UU = untreated and undried wood chips, PD = pre-treated and dried wood chips, PU = pre-treated and undried wood chips.

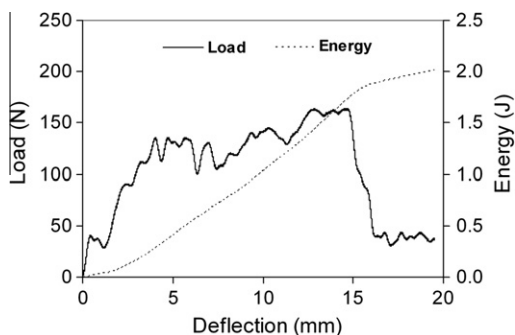


Fig. 3. Characteristic load and energy vs. deflection curves from impact testing.

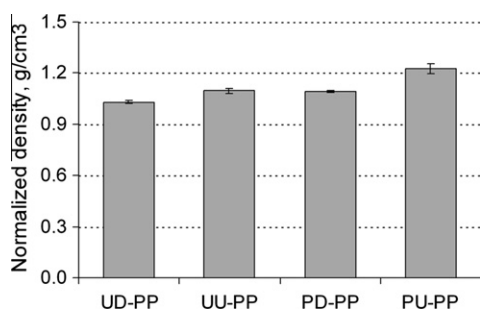


Fig. 4. Densities of the wood chip-PP composites.

3.1.3. Fracture surfaces

Figs. 5 and 6 show SEM images taken from the fracture surfaces of the wood chip-PP composites. The images in Fig. 5 are taken with less magnification to show a general view of the fracture surface of the composite. From Fig. 5 it can be seen that the fracture surface of composites with dried wood chips (Fig. 5a and c) is rougher in comparison with composites with undried wood chips (Fig. 5b and d). Also, large cavities from pull-outs and parts of large wood particles can be seen in the fracture surfaces of the compos-

ites with dried wood chips. The fracture surfaces of the composites with undried wood chips are smoother, most likely due to the smaller wood particles and fibres contained in them.

Fig. 6 shows details from the fracture surfaces of the wood chip-PP composites. It can be seen that wood particles have not been separated into individual fibres when dried wood chips have been used as raw material (Fig. 6a and c), even though the particle size appears to be greatly reduced. The presence of individual fibres in the composites with undried wood chips is obvious (Fig. 6b and d). Also, larger wood particles could be seen in the fracture surfaces of the composites with undried wood chips, but the amount was significantly lower than in the composites with dried wood chips. No obvious differences between the composites with untreated and pre-treated wood chips could be seen in the SEM images, and the adhesion appeared to be moderately good in all the composite materials. No large gaps between the matrix and the wood particles could be seen in the images; however, fibre and particle pull-outs were observed in all the composites.

3.2. Wood particles

3.2.1. Microstructure

Fig. 7 shows SEM micrographs of the wood particles after matrix removal. Individual fibres are clearly seen in the images which show the wood particles extracted from the composites manufactured with undried wood chips (Fig. 7b and d). In addition, more individual fibres can be seen in Fig. 7d in comparison with Fig. 7b, indicating that the wood chip pre-treatment improved the fibre separation from the undried wood chips. When dried wood chips have been used as raw material, the wood has remained in particulate form, as can be seen from Fig. 7a and c. There are no apparent differences between the wood particles extracted from the composites made with dried, untreated wood chips (Fig. 7a) and dried, pre-treated wood chips (Fig. 7c).

3.2.2. Aspect ratio

The average aspect ratios (l/d) as well as the average lengths and widths of the wood particles extracted from the composite samples are shown in Table 4. The aspect ratios were calculated

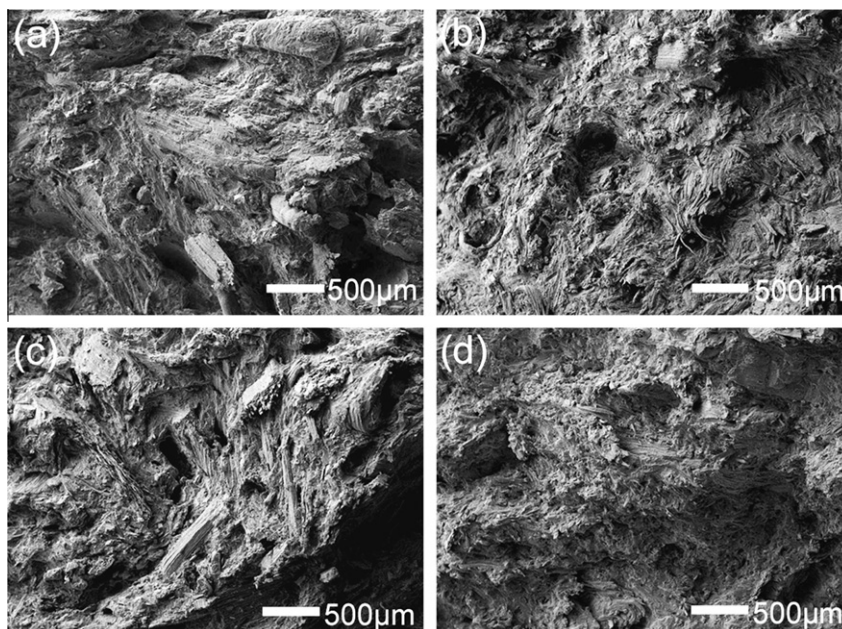


Fig. 5. Overviews of the fracture surfaces of the wood chip-PP composites. (a) UD-PP, (b) UU-PP, (c) PD-PP and (d) PU-PP.

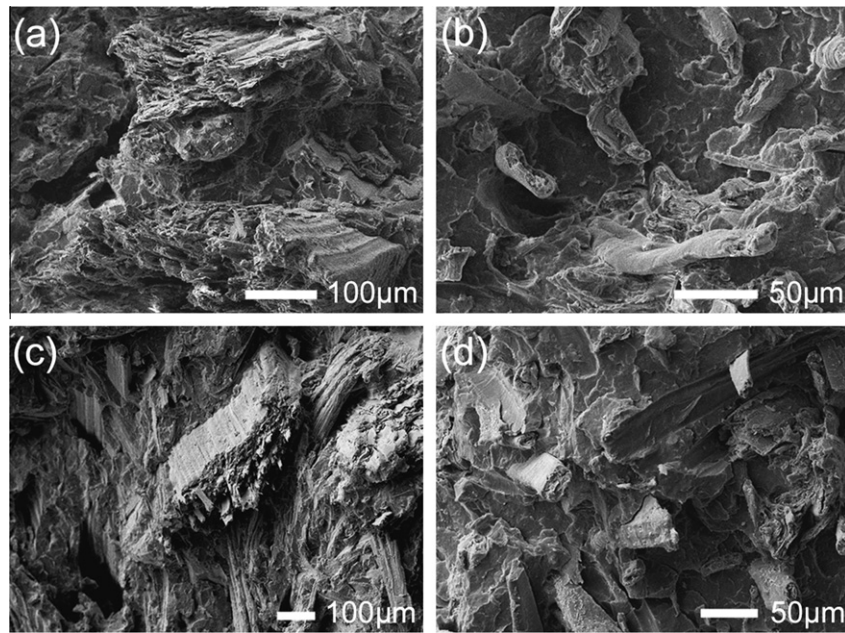


Fig. 6. Details from the fracture surfaces of the wood chip-PP composites. (a) UD-PP, (b) UU-PP, (c) PD-PP and (d) PU-PP.

Table 4

The average lengths, widths and aspect ratios of the extracted wood particles.

Material	Length, $L(l)^a$ (mm)	Width (μm)	Aspect ratio (l/d)
UD-PP	0.3	46.3	7.9
UU-PP	0.6	34.1	23.1
PD-PP	0.3	38.3	8.9
PU-PP	0.7	35.7	25.3

^a Length weighted average length.

from the length and width measurements determined using image analysis software.

As can be seen from Table 4, the wood particle samples from the composites in which dried wood chips were used (UD-PP, PD-PP) had similar aspect ratios as wood flour typically used in wood-polymer composites. The use of undried wood chips as raw material resulted in significantly greater wood particle aspect ratios after compounding. The wood chip pre-treatment slightly increased the wood particle aspect ratio, which may explain the improved mechanical properties of the composites with pre-treated wood chips. The wood particles with the highest aspect ratio were extracted from the composite made with pre-treated and undried wood chips (PU-PP), which also had the highest flexural strength and impact properties (Table 3).

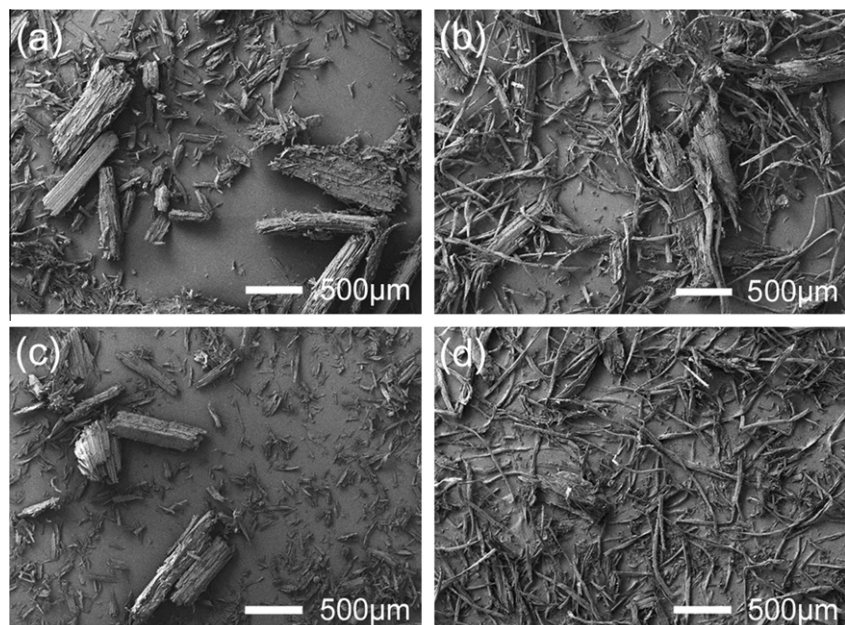


Fig. 7. Wood particles after matrix removal. (a) UD-PP, (b) UU-PP, (c) PD-PP and (d) PU-PP.

4. Conclusions

In this work the effect of chemical pre-treatment and moisture content (dried and undried wood chips) in manufacture of wood chip-PP composites was studied with an aim to create more individual wood fibres during the processing as well as improve the composites' mechanical properties. This study showed that chemical pre-treatment with sodium sulphite improved the composites' mechanical properties and that the wood particle aspect ratio was increased in the extrusion process.

The use of pre-treated wood chips enhanced the flexural properties of the wood chip-PP composites. The use of undried wood chips resulted in improved flexural strength in comparison with the composites with dried wood chips, but the flexural modulus was decreased. Study of the aspect ratios of wood particles showed that the use of pre-treated and undried wood chips resulted in the highest aspect ratio after compounding. In general, aspect ratios were considerably higher when undried wood chips were used as raw material instead of dried wood chips.

The overall results are promising. The study shows that it is possible to use wood chips as raw material for the production wood-polymer composites and that the wood chips can be defibrated during the composite manufacturing process. When a simple chemical pre-treatment is used to soften the wood a more defibrated wood structure can be achieved.

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References

- [1] Oksman K, Bengtsson M. Wood fibre thermoplastic composites: processing, properties and future developments. In: Fakirov S, Bhattacharyya C, editors. Handbook of engineering biopolymers: homopolymers, blends and composites. Munich: Carl Hanser Verlag; 2007. p. 655–71.
- [2] Clemons C. Raw materials for wood-polymer composites. In: Oksman Niska K, Sain M, editors. wood-polymer composites. Cambridge: Woodhead Publishing Limited; 2008. p. 1–22.
- [3] Stark NM. Wood fiber derived from scrap pallets used in polypropylene composites. Forest Prod J 1999;49(6):39–46.
- [4] Stark NM, Rowlands RE. Effects of wood fiber characteristics on mechanical properties of wood/polypropylene composites. Wood Fiber Sci 2003;35(2): 167–74.
- [5] Caulfield DF, Clemons C, Jacobson RE, Rowell RM. Wood thermoplastic composites. In: Rowell RM, editor. Handbook of wood chemistry and wood composites. Boca Raton: CRC Press; 2005. p. 365–78.
- [6] Dalvåg H, Klason C, Strömwall HE. The efficiency of cellulosic fillers in common thermoplastics II: filling with processing aids and coupling agents. Int J Polym Mater 1985;11(1):9–38.
- [7] Malda D, Kokta BV, Daneault C. Influence of coupling agents and treatments on the mechanical properties of cellulose fiber-polystyrene composites. J Appl Polym Sci 1989;37(3):751–75.
- [8] Raj RG, Kokta BV, Daneault C. Use of wood fibers in thermoplastics. VII. The effect of coupling agents in polyethylene-wood fiber composites. J Appl Polym Sci 1989;37(4):1089–103.
- [9] Felix JM, Gatenholm P. The nature of adhesion in composites of modified cellulose fibers and polypropylene. J Appl Polym Sci 1991;42(3):609–20.
- [10] Ichazo MN, Albano C, González J, Perera R, Candal MV. Polypropylene/wood flour composites: treatments and properties. Compos Struct 2001;54(2–3):207–14.
- [11] Zaini MJ, Fuad MYA, Ismail Z, Mansor MS, Mustafah J. The effect of filler content and size on the mechanical properties of polypropylene/oil palm wood flour composites. Polym Int 1996;40(1):51–5.
- [12] Stark NM, Berger MJ. Effect of particle size on properties of wood-flour reinforced polypropylene composites. In: Proceedings of 4th international conference on woodfiber-plastic composites, Madison; May, 1997. p. 134–43.
- [13] Chen HC, Chen TY, Hsu CH. Effects of wood particle size and mixing ratios of HDPE on the properties of the composites. Holz Roh Werkst 2006;64(3):172–7.
- [14] Bouafif H, Koubaa A, Perré P, Cloutier A. Effects of fiber characteristics on the physical and mechanical properties of wood plastic composites. Compos Part A 2009;40(12):1975–81.
- [15] Migneault S, Koubaa A, Erchiqui F, Chaala A, Englund K, Wolcott MP. Effects of processing method and fiber size on the structure and properties of wood-plastic composites. Compos Part A 2009;40(1):80–5.
- [16] Yemele MCN, Koubaa A, Cloutier A, Soullounganga P, Wolcott M. Effect of bark fiber content and size on the mechanical properties of bark/HDPE composites. Compos Part A 2010;41(1):131–7.
- [17] Renner K, Kenyó C, Móczó J, Pukánszky B. Micromechanical deformation processes in PP/wood composites: particle characteristics, adhesion, mechanisms. Compos Part A 2010;41(11):1653–61.
- [18] Balasuriya PW, Ye L, Mai Y-W. Mechanical properties of wood flake-polyethylene composites. Part I: effects of processing methods and matrix melt flow behaviour. Compos Part A 2001;32(5):619–29.
- [19] Bledzki AK, Faruk O. Wood fibre reinforced polypropylene composites: effect of fibre geometry and coupling agent on physico-mechanical properties. Appl Compos Mater 2003;10(6):365–79.
- [20] Hietala M, Niinimäki J, Oksman K. Processing of wood chip-plastic composites: effect on wood particle size, microstructure and mechanical properties. Plast Rubber Compos 2011;40(2):49–54.
- [21] Krogerus B, Fagerholm K, Löytynoja L. Analytical fractionation of pulps by tube flow. Pap Puu-Pap Tim 2003;85(4):209–13.
- [22] Laitinen O, Löytynoja L, Niinimäki J. Tube flow fractionator: a simple method for laboratory fractionation. Pap Puu-Pap Tim 2006;88(6):1–5.
- [23] Lindholm CA, Kurdin J. Chemimechanical pulping. In: Sundholm J, editor. Papermaking science and technology. Book 5. Mechanical Pulping. Jyväskylä: Fapet Oy; 1999. p. 223–49.